

Practical Electrical Engineering Interview Questions

This collection emphasizes safe, practical reasoning. In every measurement scenario, confirm ratings, use appropriately rated probes and PPE, and de-energize equipment when possible.

1. You must measure 50 V, but your meter's maximum range is 10 V. What do you do?

Use a correctly rated voltage divider or a rated high-voltage probe so the meter sees no more than 10 V. For example, a 4:1 divider gives $V_{\text{meter}} = V_{\text{in}}/5 = 10 \text{ V}$ at 50 V; account for divider tolerance and meter input impedance. Do not exceed the meter or probe voltage rating.

2. A sensor is far from the controller. Would you transmit voltage or current?

Usually use a current loop, commonly 4–20 mA. Cable resistance causes a voltage drop but the loop current remains nearly unchanged if the supply has adequate compliance voltage. The 4 mA live zero also helps detect an open circuit.

3. Two unequal resistors are in parallel. Which heats more?

They have the same voltage, so $P = V^2/R$. The lower resistance dissipates more power, provided both are within their ratings.

4. Two unequal resistors are in series. Which heats more?

They carry the same current, so $P = I^2R$. The higher resistance dissipates more power.

5. How do you measure current without opening a high-current circuit?

Use a correctly rated clamp meter (Hall-effect for DC, current transformer for AC) around one conductor only. Clamping both supply and return makes their magnetic fields cancel.

6. Why should an oscilloscope ground clip not be attached casually to a mains-referenced node?

On an earth-referenced scope, the ground clip is connected to protective earth. Clipping it to a live non-earth node can create a short circuit. Use an isolated/differential probe or an approved isolated measurement setup.

7. How do you choose a shunt resistor for current sensing?

Choose $R_{\text{shunt}} = V_{\text{sense}}/I_{\text{max}}$, then check $P = I_{\text{max}}^2 R_{\text{shunt}}$, tolerance, temperature coefficient, common-mode range, and voltage burden. A small value reduces loss but also reduces signal.

8. A 10-bit ADC uses a 3.3 V reference. What is one LSB?

Approximately $3300 \text{ mV}/2^{10} = 3.22 \text{ mV}$ per code. Noise and reference error usually limit useful resolution below this.

9. What is the cutoff frequency of a first-order RC low-pass filter?

$f_c = 1/(2\pi RC)$. At f_c , the magnitude is down by 3 dB and output amplitude is $1/\sqrt{2}$ of the low-frequency value.

10. Why place a small ceramic capacitor close to an IC supply pin?

It supplies fast transient current locally and reduces supply impedance caused by trace inductance. Pair it with bulk capacitance for slower, larger load changes.

11. A regulator output oscillates after changing its output capacitor. What do you check?

Check the datasheet's required capacitance, ESR range, placement, grounding, input bypassing, load conditions, and layout. Some regulators require ESR for loop stability; others require very low ESR.

12. What is the difference between a fuse and a circuit breaker?

A fuse opens once by melting and must be replaced; a breaker is resettable and may offer adjustable trip behavior. Neither automatically protects every semiconductor from fast fault energy unless its time-current characteristic is suitable.

13. How do you estimate a resistor's voltage rating requirement?

Check both dissipation and maximum working voltage. For a resistor, $V = \sqrt{PR}$ is only a power-based limit; the datasheet's working-voltage and overload-voltage ratings can be lower constraints.

14. Why is a flyback diode placed across a relay coil?

When coil current is interrupted, $v = L di/dt$ can become very large. A reverse-biased diode provides a current path and clamps voltage, protecting the switch; it also slows relay release.

15. How can you make a relay release faster than with a plain flyback diode?

Use a higher-voltage clamp such as a TVS, diode-plus-Zener, or an appropriate active clamp. This increases the permitted coil voltage and $|di/dt|$, but the switch must tolerate the clamp voltage and energy.

16. What is power factor, and why does it matter?

For sinusoidal systems, $PF = P/S = \cos \phi$, where P is real power and $S = V_{\text{rms}} I_{\text{rms}}$. Low PF increases current for a given real power, raising cable and equipment losses; nonlinear loads also have distortion PF.

17. How do you correct inductive power factor?

Add capacitive VARs, often with capacitor banks or active PFC, after evaluating harmonics, switching transients, and resonance. Do not size capacitors solely from a nameplate without measuring the load.

18. Why is a star ground often used in mixed analog/digital systems?

It controls shared return impedance so noisy digital or power currents do not create voltage errors in sensitive analog returns. The exact topology depends on frequency and layout; at high frequency a continuous low-impedance plane is often better.

19. What is ground bounce?

Fast changing current through package, via, and trace inductance produces $V = L di/dt$, shifting a device's local ground. Reduce it with short, wide return paths, planes, decoupling, slower edges where acceptable, and controlled simultaneous switching.

20. Why terminate a fast digital signal?

If edge rise time is short relative to trace delay, a trace behaves as a transmission line. Termination matching the characteristic impedance reduces reflections, ringing, false thresholds, and EMI.

21. How do you decide whether a PCB trace needs transmission-line treatment?

Compare rise time to propagation delay, not clock frequency alone. A common rule is to consider it when trace delay exceeds roughly one-sixth of rise time; simulate or measure when margins are tight.

22. What is the skin effect?

At higher AC frequency, current concentrates near a conductor surface, increasing effective resistance. Skin depth is $\delta = \sqrt{2\rho/(\omega\mu)}$, so conductor and plating choices matter for RF and high-current AC.

23. Why twist a differential pair?

Twisting makes both conductors see similar external interference and reduces loop area, improving common-mode noise rejection and lowering magnetic pickup. Maintain the required pair impedance when routing controlled-impedance PCB pairs.

24. What does CMRR mean for an instrumentation amplifier?

Common-mode rejection ratio measures how well equal voltage on both inputs is rejected: $\text{CMRR} = 20 \log_{10}(A_d/A_{cm})$. High CMRR helps only if source impedances and input common-mode range are also well controlled.

25. A bridge sensor output is noisy. What are likely remedies?

Use a stable or ratiometric excitation/reference, differential sensing, shielding and twisted-pair cabling, low-pass filtering, proper grounding, and an instrumentation amplifier with sufficient common-mode range.

26. Why use Kelvin connections to measure a low resistance?

Separate force and sense leads. The sense leads draw negligible current, so their voltage measurement excludes lead and contact drops from the high-current force path.

27. How do you calculate transformer turns ratio?

Ideally, $V_p/V_s = N_p/N_s$ and $I_p/I_s = N_s/N_p$. Real designs also include winding resistance, core loss, leakage inductance, regulation, insulation, and saturation limits.

28. What causes transformer core saturation?

Excess volt-seconds, low frequency at fixed voltage, DC bias, too few turns, or startup asymmetry can drive flux beyond the core's linear region. Current then rises sharply and can damage windings or switches.

29. Why must a current transformer secondary never be left open while primary current flows?

An open secondary removes the normal opposing ampere-turns, allowing dangerous high secondary voltage and core heating. Short or burden the secondary using approved procedures before disconnecting instruments.

30. What is inrush current, and how can it be limited?

It is the temporary high input current when energizing capacitors, transformers, or motors. Use an NTC thermistor, precharge resistor and relay, controlled ramp, soft starter, or appropriately rated supply and protection.

31. How do you select a MOSFET for a switching application?

Check V_{DS} margin including transients, current and SOA, $R_{DS(on)}$ at actual gate drive and temperature, switching losses, gate charge, avalanche rating, thermal path, and package parasitics.

32. What are the principal MOSFET switching losses?

Conduction loss is approximately $I^2 R_{DS(on)}$. Transition loss is roughly $P_{sw} \approx \frac{1}{2} V I (t_r + t_f) f_s$, plus gate-drive, output-capacitance, reverse-recovery, and ringing losses.

33. Why is a gate resistor used with a MOSFET?

It limits peak gate current and controls edge rate, ringing, EMI, and switching stress. Too large a value increases switching loss; separate turn-on and turn-off resistances can optimize both.

34. What is a MOSFET body diode concern in a synchronous converter?

During dead time it can conduct and then suffer reverse-recovery loss (device-dependent). Ensure adequate dead time without excessive diode conduction and verify switching-node ringing and bootstrap constraints.

35. Why can a motor draw high current at startup?

Back EMF is initially near zero, so winding current is limited mainly by resistance and inductance. As speed rises, back EMF increases and current decreases; stalled current must be tolerated or limited.

36. How do you reverse a DC motor safely?

Use an H-bridge or reversing contactor, first commanding zero torque and allowing current to decay. Provide interlock/dead time so opposite switches never conduct simultaneously, and handle regenerative energy.

37. What does an insulation-resistance test measure?

A megohmmeter applies a specified DC test voltage and measures leakage through insulation. Isolate sensitive electronics first, follow equipment procedures, and discharge the tested item afterward.

38. How do you estimate battery run time?

First estimate load power and convert through converter efficiency: $I_{bat} \approx P_{load} / (V_{bat} \eta)$. Then use usable battery capacity at that discharge rate and temperature; nominal amp-hours alone can overstate runtime.

39. Why should lithium-ion cells not be connected directly in parallel without consideration?

Unequal cell voltages can cause very high equalization current. Use matched cells, appropriate protection/BMS design, fusing where needed, and controlled precharge or manufacturer-approved assembly methods.

40. What is a 4–20 mA loop's compliance voltage?

It is the supply voltage margin needed to drive the specified current through cable resistance, receiver burden, and transmitter. Verify V_{supply} exceeds all drops plus the transmitter's minimum operating voltage.

41. How would you detect a broken 4–20 mA sensor wire?

Configure the receiver to treat current below the valid live-zero threshold (often below about 3.6 mA by convention) as a fault. Exact alarm levels must follow the transmitter and system specification.

- 42. Why is differential measurement preferred for a low-level remote sensor?**
It rejects voltage differences caused by ground offsets and coupled noise when the receiver's common-mode range and CMRR are adequate. Use twisted pair, shielding strategy, and input protection as well.
- 43. What is aliasing in data acquisition?**
Frequency content above half the sample rate folds into lower apparent frequencies. Prevent it with an analog anti-alias low-pass filter and sample rate f_s sufficiently greater than twice the highest retained frequency.
- 44. What is quantization noise?**
It is error from mapping a continuous signal to discrete ADC codes. For an ideal ADC and suitable signal, it is often modeled as uniform noise of about one LSB peak-to-peak; oversampling can help only when noise/dither and bandwidth permit.
- 45. How do you protect a microcontroller input connected to an external cable?**
Define the threat (ESD, surge, wrong voltage), then use series impedance, rail clamps or TVS at the connector, filtering, controlled return paths, and verify clamp current does not exceed the IC's limits.
- 46. A buck converter output is too low. What do you check first?**
Check input voltage under load, enable/UVLO state, feedback-divider value and routing, current limit, inductor saturation, diode/synchronous switch orientation, compensation, and load short circuits. Probe with a short ground spring to avoid misleading ringing.
- 47. Why might an inductor saturating cause a converter failure?**
After saturation inductance collapses, so current slope $di/dt = V/L$ rises sharply. This can trigger current limit, overheat a switch, or cause unstable output; select saturation current with transient margin.
- 48. How do you calculate voltage drop in a DC cable?**
For a two-wire run, $V_{\text{drop}} = I(2L)R_{\text{per length}}$. Include temperature-adjusted resistance, connector losses, startup current, allowable voltage at the load, and protection coordination.
- 49. What is the purpose of a bleeder resistor across a high-voltage capacitor?**
It discharges stored energy after power removal. The voltage decays as $V(t) = V_0 e^{-t/(RC)}$; choose R and its voltage/power rating to meet a safe discharge time while limiting continuous loss.
- 50. How much energy is stored in a capacitor, and why is it safety relevant?**
 $E = \frac{1}{2}CV^2$. Even modest capacitance at high voltage can deliver dangerous energy; verify absence of voltage with a rated meter and use approved discharge procedures rather than assuming a capacitor is safe.
- 51. How do you troubleshoot an intermittent electrical fault?**
Reproduce it safely while logging voltage, current, temperature, and signals; inspect connectors, solder joints, harness strain, grounds, and supply dips. Change one condition at a time and avoid creating a new fault with probing.